

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) has emerged as a promising framework for enhancing the reasoning capabilities of large language models. However, existing approaches such as GRPO often suffer from zero gradients. This problem arises primarily due to fixed clipping bounds for token-level probability ratios and the standardization of identical rewards, which can lead to ineffective gradient updates and underutilization of generated responses. In this work, we propose **Dynamic Clipping Policy Optimization (DCPO)**, which introduces a dynamic clipping strategy that adaptively adjusts clipping bounds based on token-specific prior probabilities to enhance token-level exploration, and a smooth advantage standardization technique that standardizes rewards across cumulative training steps to improve the response-level effective utilization of generated responses. Evaluated head-to-head against GRPO, DAPO, and GSPO across four benchmarks and four different models, DCPO delivers the strongest overall performance in every comparison, and its advantage is largest precisely on the hardest, most competitive tasks. On AIME24 with the Qwen2.5-Math-7B model, DCPO reaches an Avg@1 of 46.7 under greedy decoding and an Avg@32 of 38.8 under 32-sample decoding, leading the next-best baseline GSPO (40.0/34.9) by +6.7/+3.9 and outpacing DAPO (36.7/31.6) and GRPO (36.7/32.1) by roughly +10/+7. The same pattern holds on AIME25 with Qwen2.5-14B, where DCPO scores (23.3/19.0) and again tops every baseline—GRPO (13.3/10.5), DAPO (20.0/15.3), and GSPO (16.7/9.9)—by margins as large as +10.0/+8.5 over GRPO. Beyond accuracy, DCPO is also markedly more data-efficient: averaged over the four models it raises the fraction of responses carrying a nonzero advantage by 28 absolute points over GRPO, halves the training cost required to match DAPO, and cuts the token clipping ratio by an order of magnitude relative to both GRPO and DAPO—all while achieving the superior accuracy above. Together, these results consistently establish DCPO as a more accurate and substantially more efficient way to leverage generated data for reinforcement learning in large language models.

1 INTRODUCTION

In recent years, the reasoning capabilities of large language models (LLMs) have garnered increasing attention (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Reinforcement Learning from Verifiable Rewards (RLVR) represents a promising technique for enhancing reasoning capabilities. This method leverages rule-based outcome rewards to optimize LLMs, specifically enhancing the performance of LLMs in domains requiring reflective decision-making, such as mathematics and coding (DeepSeek-AI et al., 2024).

To support RLVR, the GRPO (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) family of algorithms was developed. Although the robust performance of GRPO has been extensively validated in numerous studies, recent research, such as DAPO (Yu et al., 2025), has identified two critical and measurable limitations. First, GRPO suffers from entropy collapse, in which advantage values decay toward zero during later training stages; as our own measurements later show, this drives the fraction of usable responses from above 90% down to below 50% (and as low as 30% on Qwen2.5-Math-7B). Second, inappropriate weight-clipping discards rarely sampled learning signals, quantifiably shrinking the diversity of model exploration.

To address these challenges, for example, DAPO introduces the Clip-Higher strategy to mitigate entropy collapse by relaxing the upper clipping bound, and the Dynamic Sampling strategy to filter out

samples with uniform rewards, ensuring that every batch contains only samples with valid gradients. However, these methods carry two notable and quantifiable costs: (i) **Response-Level Inefficiency**: dynamic sampling reduces sampling efficiency, and we later measure that DAPO must generate $3\text{--}5\times$ more responses than DCPO to reach the same number of updates, at least doubling GPU hours. (ii) **token clipping deficit**: our analysis shows that the optimal clipping bound varies across tokens, so a uniform upper bound is provably suboptimal and inflates the token clipping ratio. Although some methods avoid directly discarding zero-gradient responses, these limitations remain prevalent.

In this work, we introduce **Dynamic Clipping Policy Optimization (DCPO)**, a novel approach designed to address the key limitations of previous methods. First, we introduce a **smoothing technique that standardizes advantages across cumulative steps** to address the gradient descent problem caused by zeroing out the advantages of responses with the same reward. This method aggregates the reward distribution of all cumulatively generated responses with the current generated distribution, allowing more efficient response-level exploration (see Section 2). Second, we propose a **dynamic clipping method that adaptively adjusts the clipping bounds based on the old token-specific probabilities** to mitigate the drawbacks of the fixed clipping bounds, rather than setting different fixed upper and lower clipping bounds. Specifically, tokens with lower prior probabilities are granted wider clipping bounds, allowing more space for token-level exploration (see Section 2 and Appendix A.6). In addition, DCPO only computes the mean loss across tokens within each individual response instead of averaging over an entire batch, which preserves the relative advantage structure among responses to the same prompt and avoids the dilution effect of batch-level averaging.

Finally, we merged Math-DAPO-17k (Yu et al., 2025) with levels 3-5 of the MATH dataset (Liu et al., 2025) as the training dataset and selected four common mathematical reasoning benchmarks to evaluate performance. To make the comparison as controlled and as broad as possible, we ran experiments on four different models and evaluated DCPO side-by-side with GRPO, DAPO, and—where available—GSPO under identical training and decoding conditions. Across every model, DCPO attains the best overall score, and the gap widens on the two hardest AIME benchmarks under stochastic decoding: on AIME24-Avg@32 (32-sample decoding) DCPO-7B reaches 38.8, ahead of GRPO (32.1), DAPO (31.6), and GSPO (34.9)—a lead of +3.9 over the strongest baseline and up to +7.2 over the weakest; on AIME25-Avg@32 DCPO-14B reaches 19.0, ahead of GRPO (10.5), DAPO (15.3), and GSPO (9.9), a margin of +3.7 to +9.1. These accuracy gains do not come at an efficiency cost—quite the opposite. Averaged over the four models, DCPO raises the nonzero-advantage ratio by 28 absolute points over GRPO, matches DAPO’s accuracy at half the training cost, and shrinks the token clipping ratio by an order of magnitude relative to every baseline, so higher accuracy and higher data efficiency move together rather than trading off.

For preliminary details, including loss and advantage calculation methods for previous algorithms such as GRPO, DAPO and GSPO, see the Appendix A.2.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Clipping the probability ratio is a widely adopted technique in reinforcement learning to stabilize policy updates and prevent large, destabilizing parameter shifts during optimization. This mechanism is particularly important when applying importance sampling in off-policy settings. When importance sampling is reweighted by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$, an unbiased estimate of $\mathbb{E}_{q(x)}[f(x)\frac{p(x)}{q(x)}]$ can be obtained, but the variance of such an estimate can be significantly inflated, as described in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

To mitigate this variance inflation, prior works (e.g., Schulman et al. (2017)) constrain the probability ratio within a fixed clipping bound by hyperparameter ϵ as Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

However, a symmetric fixed clipping bound constrains the model’s exploration of the diversity space, particularly in regions where the old policy assigns low probability mass. In these scenarios, the absolute bound for decreasing probabilities is inherently limited, restricting the capacity for meaningful

policy updates. Although methods like DAPO attempt to address this issue with asymmetric clipping bounds, the fundamental limitation persists, and the token clipping ratio still trends upward over training (see Figure 1). Furthermore, GSPO discards entire responses whenever the sequence-level importance ratio exceeds its clipping bounds—more than 10% of nonzero-advantage responses in practice—wasting more informative tokens while retaining more high-variance tokens, which we later observe as an average token clipping ratio above 11% on Qwen2.5-Math-7B and above 15% on Qwen2.5-14B.

From the perspective of controlling the variance-bias, a more practical approach is to impose probability-dependent constraints, as formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3) and ensuring non-negativity of the square root term by taking its maximum with zero, we obtain a closed-form solution for $r(x)$. This yields **dynamic-adaptive clipping bounds** that vary adaptively with $q(x)$ and hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

Concretely, this dynamic-adaptive rule widens the admissible update range exactly where the old policy is least confident, enlarging exploration in low-probability regions while leaving high-confidence tokens near the conventional bound (see the quantitative *TCR* comparison in Section 5.2). Furthermore, inspired by the dual clipping method (Ye et al., 2020), we cap the maximum ratio for both positive and negative advantages at 10, preventing excessively wide bounds from destabilizing training. For detailed derivation and analysis of these dynamic clipping bounds from the importance sampling framework, please see Appendix A.5 and Appendix A.6.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

In prior methods (GRPO, DAPO, GSPO), the advantage $\hat{A}_{new,j}^i$ of the j -th response is standardized using only the rewards $R_{j=1,\dots,G}^i$ from the same prompt at step i (Equation (11), Appendix A.3). This step-local view creates two concrete failure modes. First, whenever sampling randomness makes all G rewards identical at a step, the advantage is exactly zero and the prompt contributes nothing to the update—the same mechanism that we later measure as more than half of GRPO’s responses being wasted (Section 5.3). Second, under high-entropy sampling, skewed label counts make the standardized advantage swing widely between steps and even flip sign, injecting instability into training.

At the same time, recent work (Yue et al., 2025) suggests that RLVR primarily refines a policy to adjust the likelihood of correct samples rather than fundamentally altering the existing distribution. Given these considerations, the overall reward distribution of responses to the same prompt can be considered as drawn from the same global distribution throughout the course of training, which motivates a *cumulative* standardization as Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the statistics are computed over all responses generated so far for the same prompt.

To mitigate fluctuations of both step-specific standardization ($\hat{A}_{new,j}^i$) and cumulative standardization ($\hat{A}_{total,j}^i$), we introduce two smoothing functions, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$, which represent the weighted average between the two standardization methods with the weights changing over step i in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i}\hat{A}_{new,j}^i + \frac{1}{i}\hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i}\hat{A}_{new,j}^i + \frac{i-1}{i}\hat{A}_{total,j}^i \quad (6)$$

$\hat{S}A_{new,j}^i$ places more weight on the current distribution as training progresses, reflecting a preference for adapting to the latest policy outputs. Conversely, $\hat{S}A_{total,j}^i$ places more weight on the overall cumulative distribution, viewing the deviation of the current step as a fine-tuning adjustment within the broader distribution. Finally, our final advantage $\hat{A}_j^i$ is defined as the smoothed advantage with the smaller absolute value to reduce the impact of the respective fluctuations of cumulative standardization and standardization of the current step on training stability, defined as Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

Once the prompt participates in the model optimization, the responses generated in the subsequent steps will participate in the model optimization regardless of whether the current advantage is 0 or not. Consequently, when identical rewards occur within a step, these responses will also update the model with the advantage of $\frac{1}{i}\hat{A}_{total,j}^i$, preserving useful learning signals and improving data efficiency.

3.1 OTM LOSS

Although the **Token-Level Mean loss (TLM)** (the loss Equation (13) used in DAPO) balances all token contributions, it disrupts the relative relationship between responses of different lengths, and a concrete example makes the size of the distortion explicit. Consider two responses to the same prompt: response *A* has an advantage of 1 and a length of 500, while response *B* has a lower advantage of 0.5 but a length of 1500. Under TLM, *A* is down-weighted to $\frac{500}{500+1500} = 0.25$ and *B* is up-weighted to $\frac{1500}{500+1500} = 0.75$, so their effective loss contributions become $1 \times 0.25 = 0.25$ for *A* versus $0.5 \times 0.75 = 0.375$ for *B*. In other words, *B* ends up exerting 50% more influence (0.375 vs. 0.25) than the higher-advantage response *A*, purely because it is longer—an inversion of the intended ranking that OTM is designed to remove.

At the same time, the batch-averaged **Sequence-Level Mean loss (SLM)** (the loss Equation (12) used in GRPO) weakens the magnitude of the advantage, effectively scaling it down by a factor of G after the advantage is computed. This shrinkage grows with batch size and, when the batch is large, destroys the relative correlations that the original advantage standardization was meant to preserve. We believe that standardized advantages already have relative relationships across responses, and the average across all tokens in the same response with $|o_i|$ not only maintains the original relative relationship, but also distributes the importance equally to each token in the current response. As a result, we only averaged the tokens within the one response without averaging all responses in the same batch. This is called **Only Token Mean loss (OTM)** as Equation (8).

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

As in DAPO, the loss of DCPO is not constrained by the KL divergence. For the lower and upper clipping bounds, we used the dynamic clipping bounds introduced in Section 2.

4 EXPERIMENTAL SETUP

Models and Datasets To test whether DCPO’s advantage holds across model families and scales rather than at a single operating point, we experimented with four different models spanning 1.5B to 14B parameters: Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b). All four share the same training corpus—a merged set of roughly 25k mathematical problems combining the DAPO-Math-17K corpus (Yu et al., 2025) with levels 3-5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025)—and the identical Qwen-Math template (see Section A.8) for both training and

evaluation, so that any performance differences are attributable to the optimization method rather than to data or prompting.

Evaluation We evaluated every method on the same four widely used mathematical reasoning benchmarks—MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA)—so that all comparisons are made on identical test sets. For **MATH500** we report Avg@1 only, given its larger size (500 problems); for the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) datasets we report both Avg@1 and Avg@32, pairing a single-shot and a multi-sample view so that robustness claims rest on two complementary metrics rather than one.

- **Avg@1**: single-shot accuracy under greedy decoding, capturing the model’s best single prediction.
- **Avg@32**: mean accuracy over 32 sampled responses per problem at temperature 1.0 and top-p 1.0, capturing the robustness and stability of the trained policy distribution under stochastic decoding.

Baseline Settings We compared DCPO against GRPO and DAPO on all four models, and added GSPO as a third baseline on the two larger models, Qwen2.5-Math-7B and Qwen2.5-14B, giving up to four methods under directly comparable conditions. All other settings—the clipping thresholds ϵ_* , the codebase, and the common training hyperparameters—are held fixed across methods and detailed in Appendix A.9, isolating the algorithmic differences as the source of the reported gaps.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1 places DCPO side-by-side with GRPO, DAPO, and GSPO on the four math reasoning benchmarks (MATH500, AMC23, AIME24, AIME25) under both greedy decoding (Avg@1) and 32-sample decoding (Avg@32), giving a two-metric view of accuracy and robustness for each method. The single most consistent pattern in the table is that DCPO holds the best Average score in every one of the four model blocks, under both metrics. This across-the-board win—rather than a win confined to one model or one benchmark—is what identifies DCPO as a general-purpose optimization technique for RLVR rather than a method tuned to a single setting.

Overall Superiority of DCPO Reading the Average column across the four model blocks shows a win at every scale, with the margin over baselines growing as the models get larger and stronger. On Qwen2.5-Math-1.5B-Instruct, DCPO posts the top average (47.2/32.8), ahead of GRPO by +1.2/+0.8 and DAPO by +0.7/+0.4. On Qwen2.5-3B it reaches (37.6/22.7), beating GRPO by +1.3/+1.7 and edging DAPO with a clear +1.0 in Avg@1 against a negligible -0.4 in Avg@32—essentially a tie on the multi-sample metric and a win on the single-shot one. The gaps widen on the larger models: on Qwen2.5-Math-7B, DCPO’s (57.1/45.2) leads GRPO by +4.0/+3.6, DAPO by +3.3/+2.8, and GSPO by +1.9/+1.9, i.e. it is ahead of all three baselines on both metrics simultaneously. On the largest model, Qwen2.5-14B, DCPO’s (53.2/39.0) extends the lead to +6.5/+7.7 over GRPO, +4.2/+5.5 over GSPO, and +1.3/+0.1 over DAPO. Taken together, DCPO is never behind on the Average column, and its advantage is most pronounced exactly where capacity is highest, indicating that its benefits scale with model size rather than washing out.

Robustness on AIME under Sampling (Avg@32) DCPO’s largest margins appear precisely on the hardest tasks under the most demanding metric. AIME24 and AIME25 are high-difficulty competition benchmarks, and Avg@32 stresses whether a model can produce correct answers under high-entropy sampling rather than in a single greedy shot. On the 7B and 14B models, DCPO pulls clearly ahead: on AIME24-Avg@32, DCPO-7B scores 38.8, beating the best baseline GSPO (34.9) by +3.9 and outscoring DAPO (31.6) and GRPO (32.1) by +7.2 and +6.7—a relative improvement of roughly 20% over both. On AIME25-Avg@32, DCPO-14B reaches 19.0, exceeding DAPO (15.3) by +3.7 and nearly doubling GRPO (10.5, +8.5) and GSPO (9.9, +9.1). That the advantage is widest under multi-sample decoding indicates DCPO improves not just the single best answer but the diversity and reliability of correct generations across samples—exactly the property that matters when many plausible outputs are drawn at deployment time.

5.2 THE TOKEN CLIPPING RATIO (*TCR*)

We use the **Token Clipping Ratio** (*TCR*) as the proportion of tokens excluded from policy updates during back-propagation due to clipping; a lower *TCR* means more of each generated response

Table 1: Head-to-head performance across four benchmarks and four models (Avg@1: greedy decoding; Avg@32: 32-sample decoding). Within each model block the methods share identical data and hyperparameters, so columns are directly comparable. Boldface marks the best value in each column, and the Δ rows report DCPO’s gain over each baseline (red = DCPO ahead). DCPO holds the best Average score in all four model blocks.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/ 70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

actually drives learning. As shown in Figure 1, the four methods separate cleanly on this metric, and DCPO sits well below the others throughout training.

$$TCR = Average \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } micro_m}{\text{Total number of tokens in } micro_m} \right) \quad (9)$$

where $micro_m$ denotes the m -th microbatch and N is the number of microbatches per training step.

The baselines all show unfavorable TCR dynamics, though in different ways. Under GRPO the trend is model-dependent: for the smaller models (Qwen2.5-Math-1.5B and Qwen2.5-3B) the TCR rises with training steps, whereas for the larger models (Qwen2.5-Math-7B and Qwen2.5-14B) it gradually falls. DAPO is more uniformly problematic, with the TCR climbing steadily for every model, meaning an ever-larger share of updates rests on partial or truncated responses as training proceeds. Both GRPO and DAPO also display large fluctuations and occasional abnormal spikes, a source of optimization instability that can precipitate entropy collapse. GSPO is worse still on this axis: its average TCR exceeds 11% on Qwen2.5-Math-7B and 15% on Qwen2.5-14B—several times

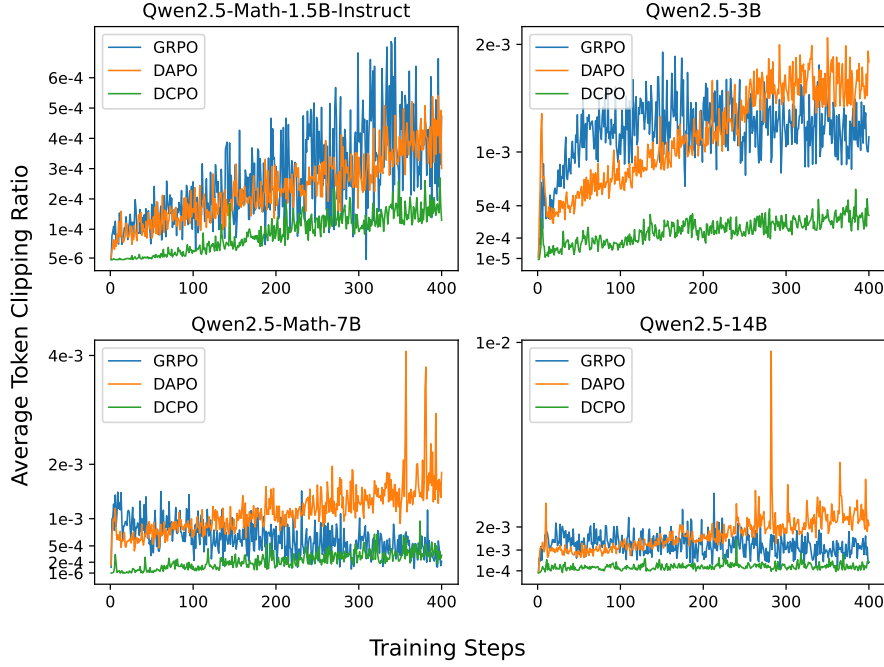


Figure 1: Token Clipping Ratio (*TCR*) over training for each method and model (lower is better). DCPO’s *TCR* stays roughly flat and about an order of magnitude below GRPO and DAPO across all four models, whereas GRPO and DAPO drift upward with recurring spikes.

the level of the token-level clipping methods—so it discards a much larger fraction of informative tokens.

DCPO stands in sharp contrast on both stability and level. Independent of model size or training stage, its *TCR* stays essentially flat and roughly an order of magnitude below GRPO and DAPO—and far below GSPO’s double-digit averages. Because so few tokens are clipped, a much larger portion of each complete response contributes to the policy update, which both improves sample efficiency and leaves more room for the model to explore rare tokens. In short, DCPO simultaneously achieves the lowest and the most stable *TCR* of any method compared here, while tightly bounding outlier variation.

Taking Qwen2.5-Math-7B as an example, the token distribution concentrates on high-confidence tokens remarkably fast: about 95% of generated tokens already have probability $q(x) > 0.9$ after only 60 steps, rising to 97% by 100 steps and continuing to climb thereafter. As illustrated in Figure 4d, once the old probability satisfies $(q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83)$, the new probability $p(x)$ participates in the update within the interval $[\frac{1}{1+\epsilon}, 1]$ and is never discarded under any clipping scheme. Since the vast majority of tokens fall in this never-clipped region, the overall *TCR* is inherently low—which is exactly what the measurements show. The remaining, and more consequential, question is what happens to the minority of low-probability tokens: Figure 1 shows that for most baselines the *TCR* grows in later training, meaning a steadily larger share of these rare tokens is clipped away, curtailing the model’s ability to explore token-level diversity.

This is why the low-probability tail matters so much. Previous work (Wang et al., 2025) showed that **high-entropy tokens**—precisely the low-probability tokens—are the primary drivers behind the emergence of reasoning ability. Because entropy and probability are inversely related, rarer tokens carry proportionally more information, so widening their admissible update range enlarges the model’s effective exploration space and directly supports the acquisition of diverse reasoning skills.

Together, the measured *TCR* behavior and this information-theoretic argument justify our **dynamic-adaptive clipping** scheme, which relaxes clipping for tokens with $q(x) < \frac{1}{1+\epsilon}$ while leaving the safe interval for high-confidence tokens untouched. By selectively widening the update interval only for

the high-entropy tail, DCPO both keeps the *TCR* low and stable and restores the model’s capacity to learn from rare, informative tokens—the combination that underlies its reasoning gains reported above.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

During RLVR training, a policy update is contingent on a nonzero gradient for a given response. We quantify the efficiency of this process with a metric defined as the **Response Utilization Ratio** (*RUR*), which is the percentage of nonzero advantage of generated responses that participate in the policy update.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

The contrast in Table 2 is stark. Averaged over the four models, GRPO’s *RUR* is only 43.8%, meaning that more than half of every batch of generated responses is thrown away and never contributes to a gradient—a direct measure of wasted exploration. GSPO fares little better at 45.6% on the two models where it applies. DCPO, by contrast, sustains an average *RUR* of 71.8%, a +28.0-point absolute gain over GRPO and a 64% relative improvement, and it does so consistently across models (67.1%–74.3%) rather than at a single favorable point.

Table 2: Average Response Utilization Ratio (*RUR*, higher is better) over 400 training steps. DCPO keeps 71.8% of responses on average—about $1.6\times$ GRPO (43.8%) and GSPO (45.6%)—and stays in a tight 67.1%–74.3% band across all four models.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 shows why the averages differ so much: GRPO’s *RUR* degrades over training for every model and type, falling from above 90% early on to below 50%, and on Qwen2.5-Math-7B collapsing to as low as 30% by 200 steps. This downward trajectory is the signature of GRPO’s step-local advantage standardization producing a growing fraction of zero gradients, i.e. exploration becoming progressively less efficient exactly when it is most needed. DCPO’s smooth advantage standardization reverses this trend: after the first epoch its *RUR* stabilizes around 70% and, unlike GRPO, edges upward with continued training rather than decaying.

Two features of DCPO’s *RUR* curve warrant explanation: the brief $\approx 90\%$ peak in the first few steps, and the fact that from the second epoch onward it settles near 70% rather than climbing to 100%. Both, we argue below, are expected and in fact reflect the SAS mechanism behaving as intended rather than any inefficiency.

First, the $\approx 90\%$ initial *RUR* is largely an artifact of the early-stage model producing malformed outputs that draw negative rewards, which transiently inflates the ratio; models quickly learn the fixed answer format, after which the focus shifts to genuine reasoning. Second, the plateau near 70% (rather than 100%) is explained by two identifiable sources of persistently zero advantage: **(i)** roughly half of the problems are easy enough that the model solves them consistently, and excluding these from updates is justified because the core capabilities they exercise need no further refinement. **(ii)** The remaining instances correspond to being consistently erroneous. This may arise from two factors: On the one hand, some problems are too challenging for the current model to solve, even with a correct ground-truth label. On the other hand, the DAPO-Math-17K dataset contains some items whose labels were converted to mislabeled integers, introducing potential “incorrect” labels. Discarding responses with zero advantage in these cases acts as a form of filtering, mitigating the risk of updating the model based on noisy or mislabeled data.

DAPO discards responses with identical labels before training, so its *RUR* is not directly comparable to GRPO’s or DCPO’s; we therefore quantify its efficiency by the responses generated and wall-clock time needed to reach the same number of updates. Even after mitigating GRPO-style low utilization by dynamically setting DAPO’s generation batch to $1\text{--}4\times$ the training batch, DAPO needs $3\text{--}5\times$ more generated responses than DCPO to complete the same number of parameter updates, translating into at least a $2\times$ increase in GPU hours—and, crucially, buys no accuracy gain for that extra cost (cf.

Table 1). The gap also worsens with scale: on Qwen2.5-14B, DAPO’s share of usable responses falls below 30% after 300 steps. In short, DAPO’s dynamic sampling is both the least data-efficient and the most expensive of the compared methods, while forgoing exploration that DCPO retains.

In summary, on the two efficiency axes measured here DCPO dominates both baselines: it roughly $1.6\times$ GRPO’s response utilization (71.8% vs. 43.8%) and matches DAPO’s accuracy while using 3–5 $\times$ fewer generated responses and about half the GPU hours. Higher utilization, lower cost, and more exploration space thus arrive together, and translate into the accuracy gains reported in Table 1.

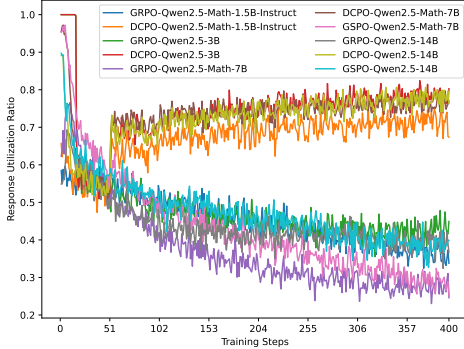


Figure 2: *RUR* over training. GRPO decays from 90% toward 50% (as low as 30% on 7B), while DCPO stabilizes near 70% and trends upward.

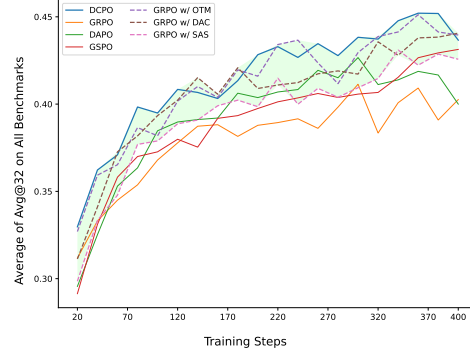


Figure 3: Component ablation on Qwen2.5-Math-7B (average Avg@32, steps 20–400). Each of OTM, SAS, and DAC individually lifts GRPO toward or past DAPO/GSPO, and the full DCPO combination is highest and keeps rising where baselines plateau.

5.4 ABLATION STUDIES

To attribute DCPO’s gains to specific design choices, we ablate its three components on Qwen2.5-Math-7B, reporting **Avg@32** to emphasize the robustness and stability of the learned policy. Each variant changes exactly one component relative to GRPO while holding all other settings fixed and removing the KL term to match DAPO, GSPO, and full DCPO, so the curves in Figure 3 (steps 20–400) isolate the effect of each mechanism.

The ablation gives a clean, monotone picture: adding any single component already lifts GRPO to the level of—or past—DAPO and GSPO, and stacking all three yields the highest curve. Just as telling is the shape late in training: the baselines flatten and stop improving, whereas both the individual DCPO components and the full model keep trending upward through step 400. Thus each mechanism contributes a distinct, positive increment, and the three combine into a larger, still-improving gain rather than saturating—evidence that the components are complementary rather than redundant. The per-component detail follows.

(i) **Only-Token-Mean loss (OTM)**: aggregating the loss within each response instead of across the batch already pushes GRPO above the baselines, which otherwise show lower data utilization and more fluctuating curves. (ii) **Smoothed Advantage Standardization (SAS)**: replacing the step-local advantage with the cumulative smoothed advantage markedly improves GRPO and brings it up to DAPO’s level, confirming that aggregating reward statistics across steps stabilizes optimization and raises final accuracy. (iii) **Dynamic Adaptive Clipping (DAC)**: swapping GRPO’s fixed clipping bound for the probability-adaptive bound is the strongest single component—DAC alone surpasses GRPO, DAPO, and GSPO, and trains more stably than OTM alone. (iv) **OTM + SAS + DAC (full DCPO)**: combining all three produces the best and most stable curve of any configuration, clearly ahead of GRPO, DAPO, and GSPO. The full model exceeding every single-component variant shows the three mechanisms are complementary contributors to data-efficient, stable RL.

Overall, the ablation attributes DCPO’s advantage to all three components acting together: each helps on its own, none is redundant, and their combination compounds into the largest and most stable gain—directly supporting the efficiency and robustness improvements documented for the full method.

6 CONCLUSION

In this work, we attempt to address key limitations of existing policy optimization methods: restricted token-level exploration due to fixed cropping in low-probability regions, and significantly low sample efficiency due to the current step’s reward standardization. To this end, we introduced **Dynamic Clipping Policy Optimization (DCPO)**, a novel reinforcement learning pipeline that enhances the reasoning capabilities of large language models through two key innovations: (i) a **dynamically adaptive token-level clipping** mechanism that adjusts bounds based on old policy probabilities, enabling more efficient exploration of rare tokens with low probabilities, and (ii) a **Cumulative-Smooth Advantage Standardization** that aggregates the cumulative reward distribution across optimization steps, mitigating reward randomness from high-entropy sampling. Across four mathematical reasoning benchmarks and four models, DCPO wins the Average score in every model block and delivers its largest margins on the hardest tasks under multi-sample decoding—e.g. on AIME24-Avg@32 the DCPO-7B variant reaches 38.8, ahead of GRPO (32.1) and DAPO (31.6) by roughly +7 (about 20% relative). These accuracy gains are accompanied by, not paid for with, efficiency gains: DCPO raises response utilization by 28 absolute points over GRPO (71.8% vs. 43.8%), matches DAPO’s accuracy at roughly half the training cost, and cuts the token clipping ratio by an order of magnitude. Higher accuracy, higher data utilization, and lower cost thus move in the same direction, underscoring DCPO’s effectiveness at improving both data utilization and diversity exploration in RLVR. Future work will extend DCPO to other domains, such as code generation and semantic reasoning.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In previous works, including GRPO and DAPO, the advantage $\hat{A}_{j,t}^i$ for the token t in the response j is calculated by standardizing the reward R_j^i against the mean μ^i and standard deviation σ^i of the rewards of the G responses generated in the i -th step, as shown in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When the rewards for the same prompt are identical, responses with zero advantages do not contribute to model update, resulting in response waste.

A.3.1 GRPO

Shao et al. (2024) proposed the **Group Relative Policy Optimization (GRPO)**, which first samples G responses for each query, assigns rewards R through a rule-based reward function and estimates token-level advantages as in Equation (11). Finally, the model parameters are updated using Equation (12), referred to as **Sequence Level Mean loss (SLM)** as:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages over all tokens in a response and then over all G responses in the batch. The $\mathbb{D}_{\text{KL}}$ penalty constrains the policy from deviating too far from a reference policy (e.g., a pre-trained model). The fixed clipping threshold ε prevents overly large updates by limiting the change in token probability ratios. This averaging strategy, though widely adopted, inherently dilutes per-response importance.

A.3.2 DAPO

Similarly, Yu et al. (2025) propose **Dynamic sAmpling Policy Optimization (DAPO)**, which observed that in GRPO, longer sequences exert a disproportionate influence on gradient update. They standardize token counts across the entire batch, leading to the **Token Level Mean loss (TLM)** as Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. It discards the prompts whose all responses share identical reward, and regenerates responses to maintain batch size, leading to severe data inefficiency.

A.3.3 GSPO

Zheng et al. (2025) propose **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping can inject high-variance noise into gradient estimates. This noise tends to accumulate over long sequences and is further amplified by the clipping operation itself. GSPO alleviates the problem by replacing token-wise clipping with sequence-wise clipping, which yields a more stable and effective training signal.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

$$\text{where } s_i(\theta) = \exp \left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log \left(\frac{\pi_{\theta}(o_{j,t}|q)}{\pi_{\theta_{\text{old}}}(o_{j,t}|q)} \right) \right)$$

In practice this sequence-level clipping discards more than 10% of responses that carry a nonzero advantage, a substantial source of data waste. GRPO, meanwhile, removes only sequence-level high-variance responses yet keeps many token-level high-entropy ones, which is a corresponding source of training instability.

A.4 REWARD CALCULATION

To provide a verifiable and interpretable supervision signal, we adopt a rule-based reward model that evaluates each generated response based on the correctness of the answers and the compliance with the format, as Equation (15). Specifically, a reward of +1 is assigned when both the format and the answer are correct; a reward of 0 is given when the format is correct but the answer is incorrect; and a reward of -1 is assigned when the format is incorrect.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j is the index of the response group for a given problem in the i -th training step, and the reward is calculated based on the format and correctness of the response.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Clipping the probability ratio is a crucial technique in reinforcement learning to stabilize policy optimization and prevent large, destabilizing parameter update. This mechanism is particularly important when using importance sampling to estimate expected values under a new policy from samples generated by an old policy.

The expected value of a function $f(x)$ under the new probability $p(x)$ can be rewritten as an expectation under the old probability $q(x)$ by importance sampling weight, as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x)\frac{p(x)}{q(x)}q(x) dx = \mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, its variance can be significantly inflated, which is a common challenge in importance sampling. The true variance of the function $f(x)$ under the new policy is given by Equation (17), whereas the variance of the importance-sampled estimator under the old policy is given by Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} [f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x)\frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The difference between the importance-sampled variance and the true variance highlights the potential for inflation, as shown in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To mitigate this difference, many prior works, including PPO Schulman et al. (2017) and RLVR methods (DeepSeek-AI et al., 2024; Liu et al., 2025), employ a fixed symmetric clipping bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

However, a fixed clipping bound inherently constrains the policy’s capacity for exploration, particularly problematic in regions where the old policy assigns a low probability mass. In such cases, the absolute bound for low probabilities is severely limited, restricting the potential for meaningful policy updates. While methods like DAPO attempt to address this by introducing asymmetric clipping bounds, the fundamental limitation of a non-adaptive constraint remains.

In addition, recent studies such as Yue et al. (2025) suggest that the reasoning paths leveraged by RLVR models are largely present within the sampling distribution of the base model. This is further evidenced by the minimal parameter shift observed during off-policy batch update, as reflected in the small fraction of clipped parameters. Given this stability, we propose a more practical alternative to constrain the probability ratio $r(x)$ through the dynamic-adaptive mechanism by including the probability in the restriction as in Equation (21), replacing the conventional fixed clipping approach of Equation (20).

$$\left| \left(\frac{p(x)}{q(x)} - 1 \right) p(x) \right| \leq \epsilon \quad (21)$$

Expanding the substitution of $p(x) = r(x)q(x)$ into the inequality of a previous equation yields the expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For this clipping ratio to be mathematically valid and meaningful, we must adhere to the fundamental constraints of probability distributions and their ratios. Specifically, as shown in Equation (23), the probabilities $p(x)$ and $q(x)$ must be non-negative and less than or equal to 1. Consequently, the probability ratio $r(x)$ must also be non-negative, and the clipping hyper-parameter ϵ must be non-negative.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To ensure the validity and practical applicability of the clipping bound derived from Equation (22), we define the necessary conditions for its bounds as outlined in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

As we can get from Equation (22) to Equation (24), the probability ratio $r(x)$ should be met by the following conditions:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For the clipping bound, we consider the overlapping region centered around the ratio value of 1 to minimize the discrepancy of bounds between the two conditions and to maintain consistency with the conventional fixed bound of GRPO, and define the following conditions to ensure the validity of the clipping bound:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, to ensure that the square number is not less than zero, we use the method of comparing with 0 and taking the maximum value of the approximation like $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. Finally, we can get the following conditions, as **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2}\sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower clipping bounds in Inequality 27, we set different upper and lower clipping thresholds ϵ_{high} and ϵ_{low} hyperparameters to align the clipping bound with the GRPO at specific points.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To ensure that our dynamic adaptive clipping bounds align with the fixed symmetric clipping bounds at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d, we derive the specific values for ϵ_{low} and ϵ_{high} by solving the equations in Equation (28).

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{\text{high}}}{q(x)}} \\ q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{\text{low}}}{q(x)}, 0)} \end{cases} \quad (28)$$

Typically, in previous practices(e.g., GRPO), the fixed clipping threshold ϵ is set to 0.2, and we can get our clipping hyperparameters to be approximately $\epsilon_{\text{low}} = 0.16$ and $\epsilon_{\text{high}} = 0.2$, while $\epsilon_{\text{low}} \neq \epsilon_{\text{high}}$ has a different purpose from the asymmetric fixed clipping setting in DAPO.

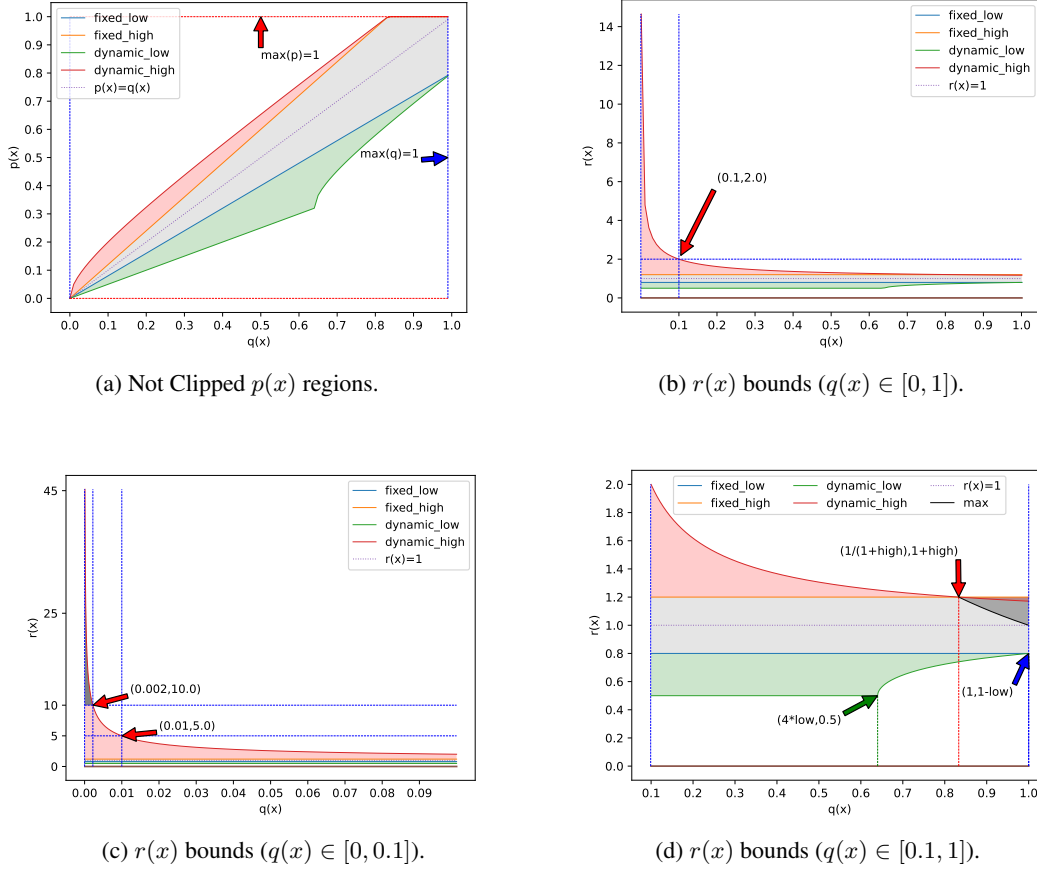


Figure 4: Clipping bound comparisons. Lines show bounds for fixed clipping ($\epsilon = 0.2$) vs. dynamic-adaptive clipping ($\epsilon_{\text{low}} = 0.16, \epsilon_{\text{high}} = 0.2$).

Dynamic versus fixed clipping: Figure 4 illustrates how the bound behaves under the fixed clipping scheme used in GRPO ($[1 - \epsilon, 1 + \epsilon]$) and under our probability-aware adaptive clipping. For a token with an old probability $q(x)$, the fixed bound provides a constant relative clipping width 2ϵ . Consequently, as $q(x)$ decreases, the absolute exploration space allowed shrinks linearly (Figure 4a). In contrast, the relative width of dynamic adaptation is inversely proportional to $q(x)$, so the exploration space remains roughly constant throughout the probability range.

Effect of decreasing $q(x)$: When $q(x)$ becomes small, the admissible interval $[1 - \varepsilon, 1 + \varepsilon]$ of the fixed clipping scheme is independent of $q(x)$, so the model’s ability to explore low-probability tokens does not improve, while the dynamic adaptive bound expands rapidly (Figure 4b), resulting in a larger relative space for rare tokens, which is exactly the behavior required for efficient RL.

Behavior for extremely low probabilities: Figure 4c shows the clipping interval when $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the upper bound for $r(x)$ of dynamic-adaptive clipping increases rapidly inversely proportional to the square root, while the corresponding value for fixed clipping remains constant. Inspired by the dual-clipping strategy (Ye et al., 2020), we enforce a hard ceiling $r_{\max} = 10$ to prevent the resulting instabilities (gradient vanishing, explosion, etc.). The gray region in Figure 4c corresponds to approximately $q(x) \leq 2 \times 10^{-3}$, where the upper limit of adaptive clipping is set to $r_{\max}$.

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\varepsilon$ the dynamic-adaptive clipping yields $dynamic_{low} = 0.5$, whereas the fixed clipping retains the constant $fixed_{low} = 1 - \varepsilon = 0.8$. When $q(x)$ increases beyond this threshold, the dynamic-adaptive lower bound smoothly contracts towards the fixed value. Similarly, the dynamic-adaptive upper bound exceeds the fixed bound whenever $q(x) < \frac{1}{1+\varepsilon_{high}}$, and merges with it for larger $q(x)$. Thus, the adaptive interval $[dynamic_{low}(q(x), \varepsilon_{low}), dynamic_{high}(q(x), \varepsilon_{high})]$ provides a probability-dependent widening of the admissible region for rare tokens while preserving the tightness of the fixed interval for common tokens (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) contains both a positive and a negative advantage term. During the operation with minimum loss value, the ratio $r(x)$ for a positive advantage is clipped in the interval $[0, dynamic_{high}(q(x), \varepsilon_{high})]$. For a negative advantage, the clipping interval $[dynamic_{low}(q(x), \varepsilon_{low}), \infty)$. Due to an excessively large $r(x)$, which can destabilize training, we limit the ratio for both positive and negative advantages at $r_{\max} = 10$. This precaution is inspired by Ye et al. (2020) and ensures that the adaptive limits do not produce excessively high importance weights.

In short, regardless of positive or negative advantages, compared to fixed clipping, dynamic adaptive clipping provides the model with more space to explore diversity when rare tokens have a low probability.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i denote the i -th iteration where prompts are used to generate responses and update the model parameters. We define the following statistics for reward normalization.

For the average for the rewards of responses, we define the equations as the group of functions as in Equation (29):

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G * (i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total} = \frac{1}{G * i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

For the variance for the rewards of the responses, we define the equations as the group of functions as in Equation (30):

$$\begin{cases}
\sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\
\sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{old}^k)^2} \\
\sigma_{total}^i = \sqrt{\frac{1}{G * i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^k)^2} \\
= \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)}
\end{cases} \quad (30)$$

To reduce resource usage, for the calculation of the value of σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29), we use equivalent calculations of μ_{new}^i , σ_{new}^i , μ_{old}^i and σ_{old}^i instead of direct calculations of original rewards.

A.8 THE TEMPLATE FOR EXPERIMENTS

We use the following template for all experiments in this paper, which is based on the Qwen-Math template (Yang et al., 2024b).

Qwen-Math Template

```

<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant

```

A.9 TRAINING SETTING

We set a symmetric clipping threshold of $\epsilon = 0.2$ in GRPO and lower/upper clipping thresholds (0.2, 0.28) with an additional 512-token soft-punish cache and the maximum probability ratio 10 for negative advantage for DAPO, lower/upper clipping thresholds ($3e - 4, 4e - 4$) for GSPO, and low/high clipping thresholds (0.16, 0.2) for DCPO, preserving the same clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1+\epsilon)$ and $(q(x), r(x)) = (1, 1-\epsilon)$ as GRPO (Figure 4d), and set the maximum probability ratio to 10 for both positive and negative advantages, different from DAPO.

Training was conducted for 400 steps with a batch size of 512 for response generation and a mini-batch size of 32 for parameter update. We used a temperature equal to 1 and top_p equal to 1 to generate $G=16$ responses for each problem, while the prompt max length was set to 1,024 and the response max length was set to 3,072, which is much smaller than the maximum length of 20k used in the original DAPO.

We adapted our training codebase from Verl (Sheng et al., 2024) on 32xH20 GPUs. Unless stated otherwise, the hyperparameters matched those of DAPO. For loss computation, we retained the original $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted average formulation from GRPO or DAPO, instead of the approximate weighted calculation used in Verl, to ensure compatibility with different microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 tracks the average Avg@1 metric—the mean greedy-decoding accuracy over MATH500, AMC23, AIME24, and AIME25—over training. Two things stand out. First, on level: DCPO’s curve stays above GRPO and DAPO for all four models, and above GSPO on the two models where it was run, throughout training rather than only at the endpoint. Second, on stability:

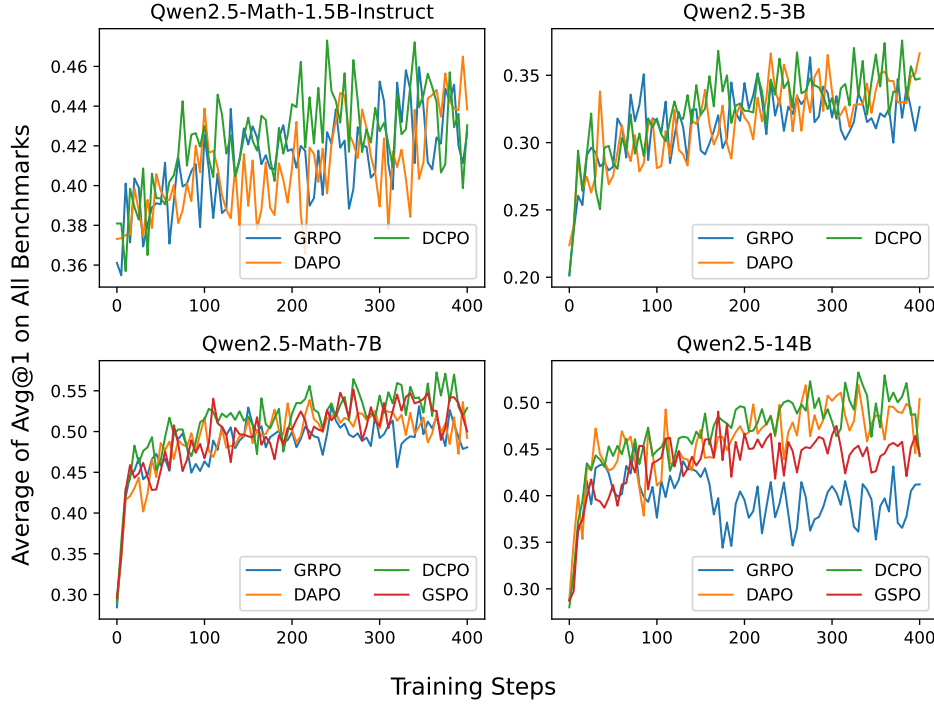


Figure 5: Avg@1 (greedy) averaged over MATH500, AMC23, AIME24, and AIME25 across training. DCPO leads GRPO and DAPO on all four models (and GSPO on the two where it is run) with a smoother, more stable trajectory.

DCPO avoids both GRPO’s pronounced mid-training drop and DAPO’s large oscillations, giving the smoothest trajectory of the compared methods. The combination of a consistently higher and steadier curve indicates DCPO elicits stronger reasoning while leaving more headroom for further improvement.

Avg@32 Trend Figure 6 compares Avg@32 for DCPO, GRPO, DAPO, and (where available) GSPO on AMC23, AIME24, and AIME25 across model sizes, and the robustness gap is visible at a glance. GRPO fluctuates heavily and occasionally collapses—on Qwen2.5-14B its Avg@32 drops sharply after 170 steps—whereas DCPO’s trajectory climbs consistently with minimal variance across the full 1.5B-to-14B range. DCPO also has the fastest early growth and never gives up its upward trend, and it reaches this superior Avg@32 while using about half the training time and resources of DAPO. Together these curves show that dynamic-adaptive clipping and smooth advantage standardization improve robustness at both the response and label levels, not just peak accuracy.

Entropy Trend Figure 7 contrasts the entropy dynamics of the three regimes. GRPO’s entropy decays sharply after a brief warm-up and settles into a flat, low band—an entropy collapse that strips the policy of randomness, shrinks response diversity, and caps further optimization.

The remaining methods separate by model type. On the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO all decline similarly. On the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, Qwen2.5-14B), however, the regimes diverge: DAPO sits at the highest entropy and trends upward with large fluctuations. GSPO tracks DCPO closely on Qwen2.5-Math-7B, matching it on both entropy and average accuracy, but on Qwen2.5-14B GSPO’s entropy fluctuates downward and its accuracy suffers accordingly. DCPO alone holds a moderate, notably smoother band—bounded between GRPO’s low-entropy floor and DAPO’s high-variance ceiling—across every base model.

These curves point to a U-shaped link between entropy and performance: too little entropy starves exploration and causes premature convergence, while too much or too erratic entropy destabilizes

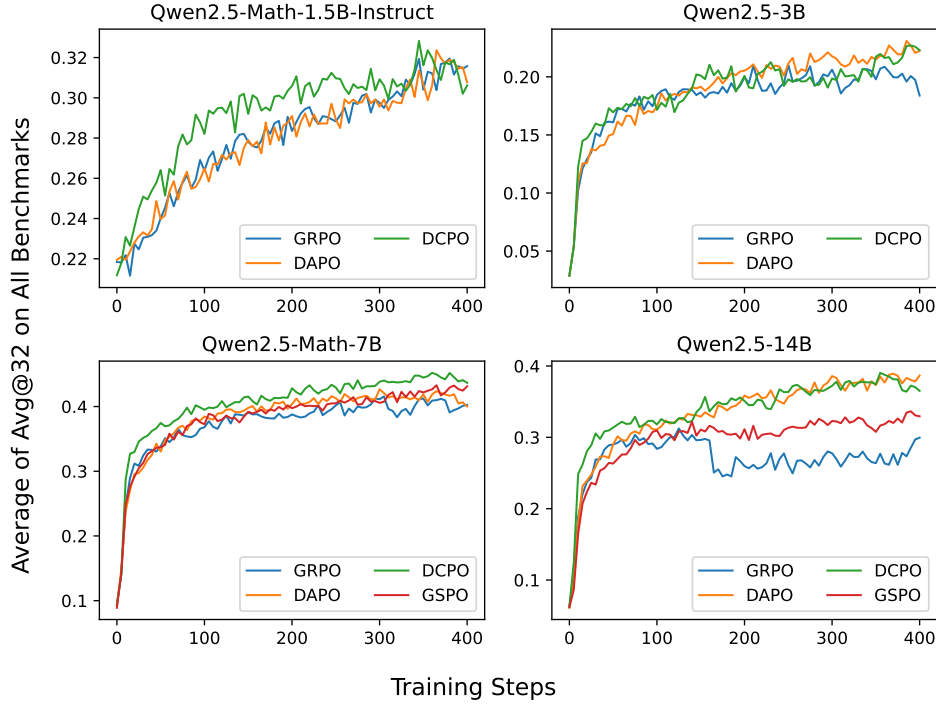


Figure 6: Avg@32 (32-sample) over training on AMC23, AIME24, and AIME25 across model sizes. DCPO rises steadily with minimal fluctuation, while GRPO fluctuates and can collapse (e.g. the 14B drop after 170 steps).

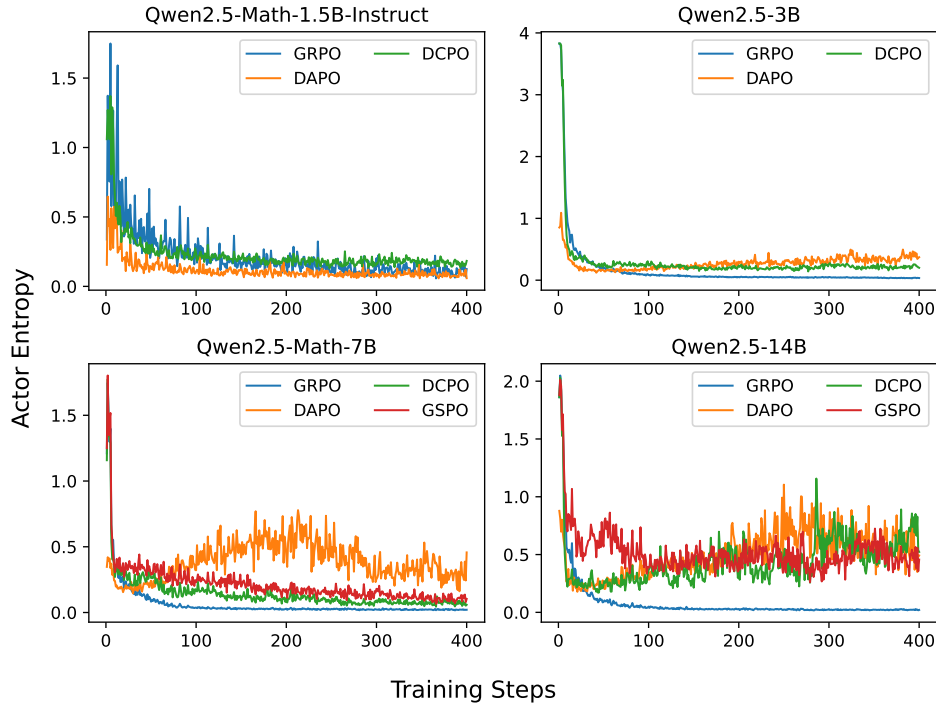


Figure 7: Policy entropy during training. GRPO collapses to a low, flat band; DAPO stays high with large fluctuations; DCPO holds an intermediate, smooth band between the two, most clearly on the larger base models.

1134 training. By keeping entropy in the balanced middle of this range—and doing so more consistently
1135 as capacity grows—DCPO secures both convergence and sufficient exploration, which plausibly
1136 explains its stronger and more robust reasoning performance across model sizes.
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